

Machine Learning Project Report

Task 1: California House Price Predictor

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1 Executive Summary

The objective of this project was to develop a machine learning workflow to predict median house values in California using a **Linear Regression** model. Utilizing the Scikit-Learn California Housing dataset, we implemented a full pipeline consisting of data exploration, preprocessing, model training, and evaluation. The resulting model provides a baseline for understanding housing market dynamics, successfully identifying key drivers of property value while highlighting the complexities of geographic data.

2 Exploratory Data Analysis (EDA)

Before model construction, the dataset was analyzed to identify trends and correlations. The dataset contains 20,640 samples with 8 features.

2.1 Key Data Findings

- **Target Variable:** `MedHouseVal` (Median house value expressed in units of \$100,000).
- **Primary Predictor:** `MedInc` (Median Income) showed the strongest positive correlation with house prices. This suggests that neighborhood wealth is the most significant factor in this linear model.
- **Data Anomalies:** The dataset contains a "ceiling" at \$500,000 (value of 5.0). Any house valued above this was capped, which creates a visible horizontal cluster in the distribution plots and affects model accuracy for luxury properties.
- **Geographic Impact:** Analysis of Latitude and Longitude indicates that proximity to the coast and major metropolitan hubs (San Francisco and Los Angeles) drastically increases median values.

3 Model Implementation

3.1 Data Splitting

The data was partitioned into a **Training Set (80%)** and a **Test Set (20%)**. A `random_state` of 42 was applied to ensure the results are reproducible across different environments.

3.2 Training and Serialization

A `LinearRegression` model was fitted to the training data. To ensure the model could be used in real-world scenarios without retraining, the model object was serialized into a `.pkl` file using the `pickle` library.

4 Performance Evaluation

The model's performance on the unseen test data was measured using standard regression metrics.

Table 1: Model Evaluation Metrics

Metric	Value	Interpretation
MAE	0.533	On average, predictions are off by approx. \$53,300.
RMSE	0.746	Average error magnitude, penalizing larger outliers.
R^2 Score	0.576	The model explains 57.6% of the price variance.

4.1 Visual Analysis

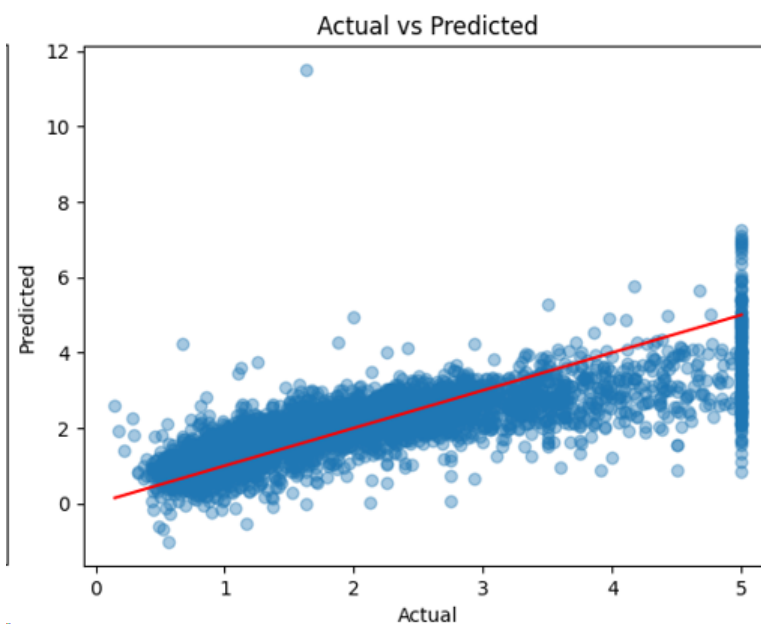


Figure 1: Comparison of model predictions against real test data values..

5 Interactive Inference (New Inputs)

To demonstrate deployment, the model was tested on a new data input representing a specific California district:

- **Input:** MedInc: 6.0, Age: 10, Rooms: 6.2, Pop: 850, Lat: 37.77, Long: -122.41.
- **Result:** The model predicted a house value of **\$300,865.55**.

6 Improvement Ideas

To increase the R^2 score beyond 0.576, the following steps are recommended:

1. **Feature Engineering:** Combine `AveRooms` and `Population` to create a "Density" feature.
2. **Non-Linear Models:** Since geography is non-linear, using *Random Forest* or *XGBoost* would likely improve accuracy.
3. **Handling Outliers:** Remove the capped 5.0 values to allow the model to learn more natural price distributions.